Levi Glenn Silvers

Curriculum Vitae

Department of Atmospheric Science
Colorado State University
Fort Collins, CO 80523
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ORCID: https://orcid.org/0000-0003-4304-2545
Google Scholar: https://tinyurl.com/ScholarSilvers

RESEARCH INTERESTS

Tropical Cyclones, predictability and internal variability of the climate, atmospheric water vapor, cloud feedback mechanisms, large-scale fluid dynamics, satellite-based observations of clouds, climate change, development of atmospheric numerical models.

EDUCATION

2011 **Ph.D.,** Atmospheric Science, Colorado State University, Fort Collins, CO.
2007 **M.S.,** Atmospheric Science, Colorado State University, Fort Collins, CO.
2004 **B.S.,** Physics, Gordon College, Minor in Mathematics, Wenham, MA.

ACADEMIC APPOINTMENTS

2024-Present **Research Scientist II,** Colorado State University, Department of Atmospheric Science, Colorado State University, Fort Collins, CO

2019-2024 **Research Scientist,** Stony Brook University, School of Marine and Atmospheric Science, Stony Brook, NY

2015-2019 **Associate Research Scholar**, Princeton University/Geophysical Fluid Dynamics Laboratory, Princeton NJ

2012-2015 **Post-Doctoral Scientist**, The Max Planck Institute for Meteorology, Hamburg, Germany

PEER REVIEWED PUBLICATIONS

32. O'Donnell, G.L., A.A. Wing, **L.G. Silvers**, K.A. Reed, P.A. Bogenschutz, J.-P. Chaboureau, G. Dagan, R. Fiévet, B. Gasparini, W.M. Hannah, K. Hashimoto, P.G. Hill, G.R. Leung, S. Matsugishi, S. Ortega, R. Roehrig, D.M. Romps, J. Shpund, A.B. Sokol, L. Tomassini, S.C.

van den Heever, C.-M. Wu; Characteristics of a Multi-model Ensemble of Mock-Walker Simulations. *J. Adv. Model. Earth Syst.,* **18,** doi:10.1029/2026MS005781.

31. Klotzbach, P.J., **L.G. Silvers,** E. Bercos-Hickey, C. J. T. Allen, M. M. Bell, E.S. Blake, N. Bloemendaal, S.G. Bowen, S. S. Chand, D.R. Chavas, M. Ekström, M. Hemmati, J.J. Jones, K.A., M.R. Lowry, C.M. Patricola, C.J. Schreck III, R.E. Truchelut, and K. M. Wood, 2026: Recent Increasing Trend in October-November Caribbean Tropical Cyclone Activity. *Geophysical Research Letters,* **53,** e2025GL120183. https://doi.org/10.1029/2025GL120183
30. **Silvers, L.G.**, P.J. Klotzbach, C.J.T. Allen, M.M. Bell, S.G. Bowen, S.S. Chand, M.Ekström, M. Hemmati, and C.J. Schreck III, 2025: The Global Influence of ENSO on Rapid Intensification of Tropical Cyclones. *Geophysical Research Letters,* **52,** e2025GL115334. https://doi.org/10.1029/2025GL115334
29. Klotzbach, P.J., E. Bercos-Hickey, K.M. Wood, C.J. Schreck III, M.M. Bell, E.S. Blake, S.G. Bowen, L.-P. Caron, D.R. Chavas, J.M. Collins, E.J. Gibney, K.A. Hansen, A.T. Hazelton, J.J. Jones, M.R. Lowry, A.T. Nieves-Jimenez, C.M. Patricola, **L.G. Silvers**, R.E. Truchelut, and J. Uehling, 2025: The Remarkable 2024 North Atlantic Mid-Season Hurricane Lull. *Geophysical Research Letters,* **52,** e2025GL116714. https://doi.org/10.1029/2025GL116714
28. Liu, P., K.A. Reed, M. Zhao,S.T. Garner, N.-C. Lau, **L.G. Silvers**, and B.A. Colle, 2025: Record-breaking persistent high-pressure systems fueled unprecedented Canadian wildfire disasters in 2023. *Environ. Res. Commun.,* **7**, 041005, doi: 10.1088/2515-7620/adc6de
27. Guo, H., **L.G. Silvers**, D. Paynter, W. Dong, S. Fan, X. Jing, R. Kramer, K. Rand, K. Suzuki, Y. Zhang, M. Zhang, 2025: Assessing clouds in GFDL's AM4.0 with different microphysical parameterizations using the satellite simulator package COSP. *Earth and Space Science.*, doi: 10.1029/2024EA004053
26. Wing, A.A., **L.G. Silvers**, K.A. Reed, 2024: RCEMIP-II: Mock-Walker Simulations as Phase II of the Radiative-Convective Equilibrium Model Intercomparison Project. *Geoscientific Model Development.,* **17,** 6195-6225, doi: 10.5194/gmd-17-6195-2024
25. **Silvers, L.G**., A. Stansfield, K.A. Reed, 2024: The impact of rotation on tropical climate, the hydrologic cycle, and climate sensitivity. *Geophysical Research Letters, **51,*** doi: 10.1029/2023GL105850
24. Thorne, L.H, T. Clay, R. Phillips, **L.G. Silvers**, E. Wakefield, 2023: A review of the effects of wind on the movement, behavior, energetics, and life history of seabirds. *Mar. Ecol. Prog. Ser.,* **173,** doi: 10.3354/meps14417
23. **Silvers, L.G**., K.A. Reed, and A.A. Wing, 2023: The response of the large-scale tropical circulation to warming. *J. Adv. Model. Earth Syst.*, **15,** doi: 10.1029/2021MS002966
22. Reed, K.A, **L.G. Silvers**, A.A. Wing, I-K. Hu and B. Medeiros, 2021: Using Radiative Convective Equilibrium to Explore Clouds and Climate in the Community Atmosphere Model. *J. Adv. Model. Earth Syst.,* **13,** doi: 10.1029/2021MS002539
21. Deroubaix, A., I. Labuhn, M. Camredon, B. Gaubert, P.-A. Monerie, et al. [including **L.G. Silvers**], 2021: Large uncertainties in trends of energy demand for heating and cooling under climate change. *Nat. Commun.* **12,** 5197 doi: 10.1038/s41467-021-25504-8
20. Burnett, A.C., A. Sheshadri, **L.G. Silvers**, and T. Robinson, 2021: Tropical Cyclone Frequency under Varying SSTs in Aquaplanet Simulations. *Geophysical Research Letters, **48,*** doi:10.1029/2020GL091980
19. Fueglistaler, S., and **L.G. Silvers**, 2021: The Peculiar Trajectory of Global Warming. *J. of Geophys. Res. Atmos.,* doi: 10.1029/2020JD033629
18. Zhang, G., **L.G. Silvers**, M. Zhao, and T.R. Knutson, 2021: Idealized Aqua-Planet Simulations of Tropical Cyclone Activity: Significance of Temperature Gradients, Hadley Circulation, and Zonal Asymmetry. *J. Atmos. Sci.,* doi: 10.1175/JAS-D-20-0079.1
17. **Silvers, L.G**., and T. Robinson, 2021: Clouds and Radiation in a mock-Walker Circulation. *J. Adv. Model. Earth Syst,* **13,** doi: 10.1029/2020MS002196
16. Wing, A.A., C.L. Stauffer, T. Becker, K.A. Reed, M.-S. Ahn, et al. [including **L. G. Silvers**], 2020: Clouds and convective self-aggregation in a multi-model ensemble of

radiative-convective equilibrium simulations. *J. Adv. Model. Earth Syst.,* **12,** doi: 10.1029/2020MS002138

15. Winton, M., A. Adcroft, J.P. Dunne, I.M. Held, E. Shevliakova, et al. [including **L. G. Silvers**], 2020: Climate sensitivity of GFDL's CM4.0. *J. Adv. Model. Earth Syst.,* **12,** doi: 10.1029/2019MS001838
14. Held, I. M., H. Guo, A. Adcroft, J. P. Dunne, L. W. Horowitz, et al. [including **L. G. Silvers**], 2019: Structure and Performance of GFDL's CM4.0 Climate Model. *J. Adv. Model. Earth Syst.,* **11,** doi: 10.1029/2019MS001829
13. Fan, S., P. Ginoux, C.J. Semen, **L.G. Silvers**, and M. Zhao, 2019: Towards improved cloud-phase simulation with a mineral dust and temperature-dependent parameterization for ice nucleation in mixed-phase clouds. *J. Atmos. Sci.* doi: 10.1175/JAS-D-18-0287.1
12. Andrews, T., J. Gregory, D. Paynter, **L. Silvers**, C. Zhou, et al., 2018: Accounting for changing temperature patterns increases historical estimates of climate sensitivity. *Geoph. Res. Lett.,* **45,** 8490-8499. doi: 10.1029/2018GL078887
11. Giorgetta, M., R. Brokopf, T. Crueger, M. Esch, S. Fiedler, et al. [including **L. G. Silvers**], 2018: ICON-A, the atmospheric component of the ICON Earth System Model. Part I: Model Description. *J. Adv. Model. Earth Syst.* **10,** doi: 10.1029/2017MS001242
10. Zhao, M., J. C., Golaz., I. M. Held, H. Guo, V. Balaji, et al. [including **L. G. Silvers**], 2018: The GFDL global atmosphere and land model AM4.0/LM4.0: 1. Simulation characteristics with prescribed SSTs. *J. Adv. Model. Earth Syst.* **10,** doi: 10.1002/2017MS001208
9. Zhao, M., J. C., Golaz., I. M. Held, H. Guo, V. Balaji, et al. [including **L. G. Silvers**], 2018: The GFDL global atmosphere and land model AM4.0/LM4.0: 2. Model description, sensitivity studies, and tuning strategies. *J. Adv. Model. Earth Syst.* **10,** doi: 10.1002/2017MS001209
8. Paynter, D., T. Frolicher, L. Horowitz, and **L.G. Silvers**, 2018: Equilibrium climate sensitivity obtained from multi-millennial runs of two GFDL climate models. *J. Geophys. Res. Atmos.*, **123,** doi:10.1002/2017JD027885.
7. **Silvers, L.G**., D. Paynter, and M. Zhao, 2018: The diversity of cloud responses to twentieth-century sea surface temperatures. *Geoph. Res. Lett.*, **45,** 391-400. doi:10.1002/2017GL075583.
6. Popp, M. and **L. G. Silvers**, 2017: Double and single ITCZs with and without clouds. *J. Climate.*, **30**, 9147-9166, doi:10.1175/JCLI-D-17-0062.1.
5. **Silvers, L. G**., B. Stevens, T. Mauritsen, and M. Giorgetta, 2016: Radiative convective equilibrium as A framework for studying the interaction between convection and its large-scale environment. *J. Adv. Model. Earth Syst.,* **8,** doi:10.1002/2016MS000629.
4. Hohenegger, C., L. Schlemmer, and **L.G. Silvers**, 2015: Coupling of convection and circulation at various resolutions. *Tellus A*, **67,** 1-17, doi:10.3402/tellusa.v67.26678
3. **Silvers, L.G**., and Wayne H. Schubert, 2012: A theory of topographically bound balanced motions and application to atmospheric low-level jets. *J. Atmos. Sci.*, **69,** 2878-2891, doi:10.1175/JAS-D-11-0309.1
2. Schubert, W.H., **L.G. Silvers**, M.T. Masarik, and A.O. Gonzalez, 2009: A filtered model of tropical wave motions. *J. Adv. Model. Earth Syst.*, **1,** Art. #3, 11pp., doi:10.1029/JAMES.2009.1.3
1. Schubert, W.H., R.K. Taft, and **L.G. Silvers**, 2009: Shallow water quasi-geostrophic theory on the sphere. *J. Adv. Model. Earth Syst.,* **1,** Art. #2, 17pp., doi:10.1029/JAMES.2009.1.2

SUBMITTED OR IN PRESS

Klotzbach, P.J., M. M. Bell, S.G. Bowen, E. Bercos-Hickey, D.R. Chavas, S. B. Girimurugan, M. Hemmati, M. R. Lowry, J. Muller, C.M. Patricola-DiRosario, C.J. Schreck III, L. G. Silvers, and K. M. Wood: Modulation of Direct Continental US Tropical Cyclone Fatalities by ENSO and the Atlantic Meridional Mode. *In Review for Geophysical Research Letters*

Non-PEER REVIEWED PUBLICATIONS

Klotzbach, P.J., and Silvers, L.G., The Influence of El Niño and La Niña on the Rapid Intensification of Tropical Cyclones, *Natural Catastrophy and Climate Report: 2025,* In Focus Section. Gallagher Re, January 2026.

Klotzbach, P.J., Bell, M.M., and Silvers, L.G., Tropical Cyclone Review, *Natural Catastrophe and Climate Report: 2024 – Data, Insights, and Perspectives,* Guest Section. Gallagher Re, January 2025.

Klotzbach, P.J., Bell, M.M, Silvers, L.G., Lee, J., Cólon Burgos, D., and Mesa, N., Extended Range Forecast of Atlantic Seasonal Hurricane Activity and Landfall Strike Probability for 2026., April 9, 2026.

Klotzbach, P.J., Bell, M.M, and Silvers, L.G., Extended Range Forecast of Atlantic Seasonal Hurricane Activity and Landfall Strike Probability for 2025, April 3, 2025.

Klotzbach, P.J., Bell, M.M, DesRosiers, A.J., and Silvers, L.G., Extended Range Forecast of Atlantic Seasonal Hurricane Activity and Landfall Strike Probability for 2024, April 4, 2024.

RESEARCH FUNDING

Title: *Understanding the links between tropical cyclones and tropical circulation under climate change through idealized coupled climate modeling*
Co-I; National Science Foundation, Climate & Large-Scale Dynamics Division, $677,761. (2023-2026).

Title: *Future projections of extreme heat events in SPEAR ensemble simulations*
Co-I; National Oceanic and Atmospheric Administration, Climate Program Modeling, Analysis, Prediction, and Projections (MAPP) Program, $502,292. (2023-2026).

DISSERTATION AND THESIS

Silvers, L.G., 2011: A theory of topographically bound balanced motions and application to atmospheric low-level jets. Ph.D. dissertation, Department of Atmospheric Science, Colorado State University, 78 pages.

Silvers, L.G., 2007: The theory of stratified, quasi-balanced flows on the sphere. M.S. thesis, Department of Atmospheric Science, Colorado State University, 57 pages.

INVITED TALKS

2026, Presentation, Ai2 Climate Emulator (ACE) Workshop, Colorado State University, May: *Untangling Intraseasonal Variability of Atlantic Tropical Cyclones with ACE2*
2025, Presentation, Hurricane Season Preparedness Event, United Nations Office for the Coordination of Humanitarian Affairs (UNOCHA), May: *Scientific Modelling in Action: Colorado State University's Forecast for the 2025 Season*
2024, Seminar, Woods Hole Oceanographic Institution, January: *Clouds, Water Vapor, and the Large-Scale Circulations in the Earth System*
2023, Colloquium, Swarthmore College, PA, January: *Decomposing the Links Between Clouds and the Large-scale Tropical Circulation*
2020, Seminar, University of Wyoming, WY, November: *Clouds and Climate Sensitivities Across a Hierarchy of GFDL CMIP6 Models*
2020, Seminar, Florida State University, FL, January: *A mock-Walker Circulation and Radiative Convective Equilibrium: Clouds and Precipitation*
2019, Seminar, Caltech, Climate Modeling Alliance, CA, August: *Cloud Resolving and General Circulation Model Simulations of an Idealized Walker Circulation*
2017, Seminar, McGill University, Montreal, Canada, October: *The Diversity of Cloud Responses to Twentieth-Century Sea Surface Temperatures*
2017, Seminar, Stony Brook University School of Marine and Atmospheric Science, Stony Brook, NY, March: *Variability of the Climate Feedback Parameter in AM2.1, AM3, and AM4g10r8*
2017, COSP (CFMIP Observation Simulator Package) Workshop, Institut Pierre Simon Laplace Climate Modelling Centre, Paris, France, February: *COSP at GFDL*
2016, Colloquium, Courant Institute of Mathematical Sciences, Center for Atmosphere Ocean Science, New York University, New York, NY, October: *RCE as A Framework for Studying the Interaction Between Conv. and its Large-Scale Envi.*
2016, Science Workshop Seminar, Bennington College, Bennington, VT, April: *How to Make a Computer Model of the Earth System: Theoretical Physics Meets Practical Predictive Science*

PUBLISHED DATASETS

As the lead scientist at the Geophysical Fluid Dynamics Laboratory (GFDL) for the Cloud Feedback Model Intercomparison Project (CFMIP), I created or was a co-creator for 48 datasets that have been published as part of the sixth phase of the Coupled Model Intercomparison Project (CMIP6). Details are given for 20 of these datasets below. All datasets are hosted by the Earth System Grid Federation and can be found here: https://search.datacite.org/repositories/dkrz.esgf

Silvers, L.G., C. Blanton, C. McHugh, J.G. John, A. Radhakrishnan, et al., 2018a: NOAA-GFDL GFDL-CM4 model output prepared for CMIP6 CFMIP. Earth System Grid Federation. doi:https://doi.org/10.22033/ESGF/CMIP6.1641

Silvers, L.G., C. Blanton, C. McHugh, J.G. John, A. Radhakrishnan, et al., 2018b: NOAA-GFDL GFDL-CM4 model output prepared for CMIP6 CFMIP amip-4xCO2. Earth System Grid Federation. doi:https://doi.org/10.22033/ESGF/CMIP6.8499

Silvers, L.G., C. Blanton, C. McHugh, J.G. John, A. Radhakrishnan, et al., 2018c: NOAA-GFDL GFDL-CM4 model output prepared for CMIP6 CFMIP amip-future4K. Earth System Grid Federation. doi:https://doi.org/10.22033/ESGF/CMIP6.8503

Silvers, L.G., C. Blanton, C. McHugh, J.G. John, A. Radhakrishnan, et al., 2018d: NOAA-GFDL GFDL-CM4 model output prepared for CMIP6 CFMIP amip-lwoff. Earth System Grid Federation. doi:https://doi.org/10.22033/ESGF/CMIP6.8506

Silvers, L.G., C. Blanton, C. McHugh, J.G. John, A. Radhakrishnan, et al., 2018e: NOAA-GFDL GFDL-CM4 model output prepared for CMIP6 CFMIP amip-m4K. Earth System Grid Federation. doi:https://doi.org/10.22033/ESGF/CMIP6.8507

Silvers, L.G., C. Blanton, C. McHugh, J.G. John, A. Radhakrishnan, et al., 2018f: NOAA-GFDL GFDL-CM4 model output prepared for CMIP6 CFMIP amip-p4K. Earth System Grid Federation. doi:https://doi.org/10.22033/ESGF/CMIP6.8508

Silvers, L.G., C. Blanton, C. McHugh, J.G. John, A. Radhakrishnan, et al., 2018g: NOAA-GFDL GFDL-CM4 model output prepared for CMIP6 CFMIP amip-p4K-lwoff. Earth System Grid Federation. doi:https://doi.org/10.22033/ESGF/CMIP6.8509

Silvers, L.G., C. Blanton, C. McHugh, J.G. John, A. Radhakrishnan, et al., 2018h: NOAA-GFDL GFDL-CM4 model output prepared for CMIP6 CFMIP aqua-4xCO2. Earth System Grid Federation. doi:https://doi.org/10.22033/ESGF/CMIP6.8511

Silvers, L.G., C. Blanton, C. McHugh, J.G. John, A. Radhakrishnan, et al., 2018i: NOAA-GFDL GFDL-CM4 model output prepared for CMIP6 CFMIP aqua-control. Earth System Grid Federation. doi:https://doi.org/10.22033/ESGF/CMIP6.8512

Silvers, L.G., C. Blanton, C. McHugh, J.G. John, A. Radhakrishnan, et al., 2018j: NOAA-GFDL GFDL-CM4 model output prepared for CMIP6 CFMIP aqua-control-lwoff. Earth System Grid Federation. doi: https://doi.org/10.22033/ESGF/CMIP6.8513

Silvers, L.G., C. Blanton, C. McHugh, J.G. John, A. Radhakrishnan, et al., 2018k: NOAA-GFDL GFDL-CM4 model output prepared for CMIP6 CFMIP aqua-p4K. Earth System Grid Federation. doi:https://doi.org/10.22033/ESGF/CMIP6.8514

Zhao, M., C. Golaz, I. Held, H. Guo, V. Balaji, et al. [including L.G. Silvers], 2018a: NOAA-GFDL GFDL-AM4 model output. Earth System Grid Federation. doi:https://doi.org/10.22033/ESGF/CMIP6.1401

Zhao, M., C. Golaz, I. Held, H. Guo, V. Balaji, et al. [including L.G. Silvers], 2018b: NOAA-GFDL GFDL-AM4 model output prepared for CMIP6 CMIP amip. Earth System Grid Federation. doi:https://doi.org/10.22033/ESGF/CMIP6.8493

Guo, H., J.G. John, C. Blanton, C. McHugh, S. Nokonov, et al. [including L.G. Silvers], 2018a: NOAA-GFDL GFDL-CM4 model output. Earth System Grid Federation. doi:https://doi.org/10.22033/ESGF/CMIP6.1402

Guo, H., J.G. John, C. Blanton, C. McHugh, S. Nokonov, et al. [including L.G. Silvers], 2018b: NOAA-GFDL GFDL-CM4 model output 1pctCO2. Earth System Grid Federation. doi:https://doi.org/10.22033/ESGF/CMIP6.8470

Guo, H., J.G. John, C. Blanton, C. McHugh, S. Nokonov, et al. [including L.G. Silvers], 2018c: NOAA-GFDL GFDL-CM4 model output abrupt-4xCO2. Earth System Grid Federation. doi:https://doi.org/10.22033/ESGF/CMIP6.8486

Guo, H., J.G. John, C. Blanton, C. McHugh, S. Nokonov, et al. [including L.G. Silvers], 2018d: NOAA-GFDL GFDL-CM4 model output amip. Earth System Grid Federation. doi:https://doi.org/10.22033/ESGF/CMIP6.8494

Guo, H., J.G. John, C. Blanton, C. McHugh, S. Nokonov, et al. [including L.G. Silvers], 2018e: NOAA-GFDL GFDL-CM4 model output historical. Earth System Grid Federation. doi:https://doi.org/10.22033/ESGF/CMIP6.8594

Guo, H., J.G. John, C. Blanton, C. McHugh, S. Nokonov, et al. [including L.G. Silvers], 2018f: NOAA-GFDL GFDL-CM4 model output piControl. Earth System Grid Federation. doi:https://doi.org/10.22033/ESGF/CMIP6.8666

Krasting, J.P., J.G. John, C. Blanton, C. McHugh, S. Nikonov, et al. [including L.G. Silvers], 2018: NOAA-GFDL GFDL-ESM4 model output prepared for CMIP6 CMIP. Earth System Grid Federation. doi:https://doi.org/10.22033/ESGF/CMIP6.1407

SELECT MEDIA APPEARANCES

CBS News, Atlantic hurricane season will be above average, researchers predict, April 3rd, 2025
Newsweek, Map Shows Florida Chances of Major Hurricane in 2025, April 3rd, 2025
The Coloradoan, CSU forecasters call for above-average hurricane season in Atlantic a year after worst ever, April 4th, 2025
The Coloradoan, CSU hurricane forecaster says conditions are ripe for more powerful storms like Helene, Oct. 4th, 2024
The Economist, America could face its most active hurricane season ever, May 30th, 2024
Fox Weather, live broadcast, Nov. 13th, 2024, June 11th, 2025, Nov 5th, 2025
Associated Press, A third November storm, Sara, serves notice that a busy hurricane season isn't over yet, Nov. 15th, 2024

TEACHING EXPERIENCE

Colorado State University

Instructor
Introduction to Weather and Climate (ATS 350, Fall 2025)

Teaching Assistant
Atmospheric Dynamics (ATS 601, Fall 2009)
General Circulation of the Atmosphere (ATS 605, Spring 2008)
Objective Analysis in the Atmospheric Sciences (ATS 655, Spring 2007)

Stony Brook University

Instructor
Global Atmospheric Change (ATM 305, Spring 2023)
Extreme Weather (ATM 103, Fall 2021)

Guest Lectures
Global Atmospheric Change (ATM 305, Spring 2020)
Extreme Weather (ATM 103, Fall 2019)

Princeton University

Instructor with Prison Teaching Initiative
Pre-Calculus, (Fall 2018)
Algebra, (Fall 2017, Spring 2018)

Guest Lecturer
Earth's Atmosphere, (Fall 2018)

Other Institutions

Modeling Mentor, (August 2012)
Dynamical Core Model Intercomparison Project, Summer School, National Center for Atmospheric Research, Boulder, CO

Physics Tutor, (Fall 2001,Spring 2002, Fall 2002)
Gordon College, Wenham, MA

EXPERIENCE AS A RESEARCH MENTOR

Timothy Park, Half Hollow Hills High School East 2022
Project on the frequency of the El-Niño Southern Oscillation
Samuel Park, Half Hollow Hills High School East 2021
Project on tropical cyclones, submitted to Regeneron STS
Juho Iipponen, Princeton University 2018
PhD candidate advised by Leo Donner
Project on an analytic, convectively driven Walker Circulation
Cara Tabor, Colorado State University 2009
Undergraduate Student in Physics department
Laboratory project on balanced vortices in rotating fluid

CONFERENCES, SEMINARS, WORKSHIPS, SUMMER SCHOOLS

2026, Presentation, 37th Conference on Hurricanes and Tropical Meteorology, AMS, San Diego, CA
2025, Presentation, CFMIP CloudSense Meeting, Exeter, UK
2025, Forecast Briefing 2025 Atlantic Hurricane Season, United Way Worldwide, virtual meeting

2024, Presentation, American Geophysical Union, Washington, D.C.
2023, Presentation, American Geophysical Union, San Francisco, CA
2023, Presentation, Northeast Tropical Workshop, Albany, NY
2023, Presentation, American Meteorological Society, Annual Meeting, Denver, CO
2022, Presentation, American Geophysical Union, Chicago, IL
2022, Presentation, CFMIP Meeting on Clouds, Precipitation, Circulation, and Climate Sensitivity, Seattle, WA
2022, Presentation, European Geophysical Union, virtual meeting
2022, Presentation, CESM Winter Working Group Meeting, virtual meeting
2021, Presentation, American Geophysical Union, New Orleans, LA
2021, Presentation, CFMIP Meeting on Clouds, Precipitation, Circulation, and Climate Sensitivity, virtual meeting
2020, Presentation, American Geophysical Union, virtual meeting
2020, Presentation, American Meteorological Society, Annual Meeting, Boston, MA
2019, Presentation, American Geophysical Union, San Francisco, CA
2019, Presentation, CFMIP Meeting on Clouds, Precipitation, Circulation, and Climate Sensitivity, Mykonos, Greece
2019, Presentation, International Union of Geodesy and Geophysics, Montreal, CA
2018, Presentation, American Geophysical Union, Washington, D.C.
2018, Presentation, CFMIP Meeting on Clouds, Precipitation, Circulation, and Climate Sensitivity, National Center for Atmospheric Research, Boulder, CO
2018, Princeton Research Day, Princeton University, Princeton, NJ
2018, Poster Expo, Geophysical Fluid Dynamics Laboratory, Princeton, NJ
2017, Presentation, American Geophysical Union, New Orleans, LA
2017, Presentation, CFMIP Meeting on Cloud Processes, Circulation and Climate Sensitivity, University of Tokyo, Tokyo, Japan
2017, Seminar, Geophysical Fluid Dynamics Laboratory, Princeton, NJ
2016, Presentation CFMIP/WCRP Conference on Cloud Processes, Circulation, and Climate Sensitivity, International Centre for Theoretical Physics, Trieste, Italy
2016, Poster Expo, Geophysical Fluid Dynamics Laboratory, Princeton, NJ
2015, Presentation, American Geophysical Union, San Francisco, CA
2015, Presentation, European Geophysical Union, Vienna, Austria
2015, Presentation, ICON Model Development Team Meeting, Hamburg, Germany
2015, Seminar, Geophysical Fluid Dynamics Laboratory, Princeton, NJ
2014, Presentation, American Geophysical Union, San Francisco, CA
2014, Presentation, European Geophysical Union, Vienna, Austria
2014, Klima Campus Seminar, Max Planck Institute for Meteorology, Hamburg, Germany
2013, Presentation, ICON Model Development Team Meeting, Offenbach, Germany
2013, Presentation, Interdisciplinary Conference of Young Earth System Scientists, Hamburg, Germany
2013, Presentation, ICON Model Development Team Meeting, Katlenburg, Germany
2013, Participant, European Centre for Medium-Range Weather Forecasts Training Course, Reading, United Kingdom
2012, Presentation, ICON Model Development Team Meeting, Hamburg, Germany
2012, Mentor, Dynamical Core Model Intercomparison Project, Summer School, National Center for Atmospheric Research, Boulder, CO
2012, Presentation, ICON Model Development Team Meeting, Katlenburg, Germany
2011, Presentation, American Geophysical Union, San Francisco, CA
2011, Presentation, 10th Team Meeting for the Center for Multiscale Modeling of Atmospheric Processes (CMMAP), Berkeley, CA
2010, Participant, Summer School on Atmospheric Modeling, Boulder, CO

2009, Presentation, AMS 17th Conference on Atmospheric and Oceanic Fluid Dynamics and 15th Conference on Middle Atmosphere, Stowe, VT
2008, Participant, Workshop on Isentropic Vertical Coordinates, NCAR, Boulder, CO
2006, Participant**,** Summer School on Modern Mathematical Methods in Physical Oceanography. Breckenridge, CO

SERVICE AND OUTREACH

CIRA/ATS Mentoring Program (CAMP) Fall, 2025
Colorado State University mentoring program for new graduate students to provide support and connections

Member of SoMAS Diversity Equity & Inclusion Committee 2022-2024

Presentations

Clouds and Hurricane Forecasting CO, 2025
Presentation to Highschool students for FORTCAST

Forecasting the 2025 Atlantic Hurricane Season CO, 2025
Colorado Science and Engineering Fair (high school)

Clouds: Perennial Inspiration for Artists and Scientists alike NY, 2022
übergeek Brewing Company, Riverhead, NY

Lecture on Weather and Climate NJ, 2018
YingHua International School, Kingston, NJ

Science Fair Judge
Long Island Science and Engineering Fair, (LISEF) NY, 2022
Monmouth Junction Elementary School, Monmouth Junction, NJ, 2016
Frontier Academy, Greeley CO, 2009

Session Chair,
AMS Schubert Symposium, Atmospheric Dynamics and Numerical Methods MA, 2020
Princeton Research Day, Long Ago and Far Away NJ, 2018

Judge, Outstanding Student Paper Awards, American Geophysical Union

Proposal Review
Panel, NASA ROSES (Research Opportunities in Space and Earth Science)
Panel, DOE BER (Office of Biological & Environmental Research)
Individual, NSF (Climate and Large-scale Dynamics Program)

Manuscript Peer-Review
Atmosphere
Climate and Atmospheric Science – NPJ
Geoscientific Model Development
Journal of Advances in Modeling Earth Systems
Journal of the Atmospheric Sciences
Journal of Geophysical Research

Nature Geoscience
Quarterly Journal of the Royal Meteorological Society

Internal Manuscript Review

The Geophysical Fluid Dynamics Laboratory
The Max Planck Institute for Meteorology

PROFESSIONAL AFFILIATIONS

American Geophysical Union
American Meteorological Society

PRE-PhD RESEARCH EXPERIENCE:

Research Assistant, Fall 2007- Fall 2011
Colorado State University, Fort Collins, CO
Ph.D. research on large-scale potential vorticity dynamics
Advisor: Prof. Wayne H. Schubert

Research Assistant, Spring 2005- Fall 2007
Colorado State University, Fort Collins, CO
M.S. research on global quasi-geostrophic theory

Research Assistant, Fall 2004-Spring 2005
Colorado State University, Fort Collins, CO
Worked with Prof. Henk Dijkstra on research concerning the thermohaline ocean circulation and seasonal cycles

Research Assistant, Summer 2002
University of Idaho, Moscow, ID
Research Experience for Undergraduates summer program
-Assisted Prof. David McIlroy with research in solid state physics